

Fahimeh Orvati Nia

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Education

Ph.D., Computer Engineering (In Progress)

Jan 2025 - Present

Texas A&M University, College Station, TX

Research Areas: AI, Computer Vision, Agricultural Imaging, Graph Neural Networks

Affiliated with Texas A&M AgriLife Research Institute

M.Sc., Electrical Engineering

Aug 2023 - Jan 2025

University of Notre Dame, IN, USA

GPA: 3.74

Thesis: Time-series anomaly detection using GANs and pedestrian intention prediction with GNNs

B.Sc., Electrical Engineering

Sept 2016 - Jan 2020

Amirkabir University of Technology, Tehran, Iran

Work Experience

Research Assistant, Texas A&M AgriLife Research

Jan 2025 - Present

- Computer vision and AI-based analysis of agricultural multispectral imagery.
- Plant growth analysis and phenotyping using deep learning.
- Integrating multispectral imaging with low-field MRI for comprehensive crop analysis.

Teaching Assistant, University of Notre Dame

Fall 2023 - Spring 2024

- Courses: Introduction to Electrical Engineering; Signals and Systems.
- Developed course materials and supported students with assignments and projects.

Research Assistant, University of Tehran

Jan 2022 - June 2023

- Machine learning for classification and clustering of musical instruments.

IoT Developer, NOICT Company (INRA)

July 2021 - Oct 2021

- Wireless communication between IoT devices using the Zigbee protocol.

Teaching Assistant, Amirkabir University of Technology

Sept 2020

- Guided students in designing control systems using MATLAB/Simulink.
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Skills

Languages & APIs: Python, C++, MATLAB, LaTeX

ML & CV Libraries: PyTorch, TensorFlow, Keras, OpenCV

Developer Tools: Git, Docker

Embedded Systems: STM32, Raspberry Pi, Arduino

Papers

- **Fahimeh Orvati Nia**, Hai Lin. *Traffic-aware pedestrian intention prediction*. American Control Conference (ACC), 2025. arXiv:2507.12433
Proposed a Traffic-Aware Spatio-Temporal GCN (TA-STGCN) that integrates dynamic traffic-signal states and

contextual cues for pedestrian intention prediction. Achieved improved accuracy on the PIE dataset (4.75% over a baseline).

- **Fahimeh Orvati Nia**, Joshua Peeples, et al. *Neighborhood Feature Pooling for Remote Sensing Image Classification*. WACV 2026 Workshop on Earth Observation. arXiv:2510.25077
Introduced a neighborhood feature pooling layer to capture local relationships and aggregate texture-like similarities across feature dimensions. Demonstrated consistent accuracy gains across datasets and backbones with minimal parameter overhead.
 - **Fahimeh Orvati Nia**, S. Salehi, J. Peeples. *Evaluating GAN-LSTM for Smart Meter Anomaly Detection in Power Systems*. TPEC 2026. arXiv:2601.09701
Conducted a power-system-oriented evaluation of GAN-LSTM anomaly detection on AMI time-series using the LEAD dataset (406 buildings, 1 year). Standardized preprocessing/windowing/thresholding and showed strong detection performance ($F1 = 0.89$) versus common baselines.
 - A. McFarland, L. Rossi, F. Nia, J. Peeples, A. Svyaneck. *Integrating Multispectral Imaging and Low-Field Magnetic Resonance Imaging for Comprehensive Phenotyping of Horticultural Crops*. HORTSCIENCE 60 (7), 2025.
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Selected Projects

Traffic-Aware Pedestrian Intention Prediction 2025

Traffic-aware graph neural network architecture (TA-STGCN) for pedestrian intention prediction in dynamic traffic environments; evaluated on PIE dataset.

Agricultural Imaging and Phenotyping 2025 - Present

Computer vision and AI methods for analyzing multispectral imagery of horticultural crops; integrating multispectral imaging with low-field MRI for comprehensive phenotyping.

GAN-Based Anomaly Detection 2024

Designed GAN-based models with custom loss functions for time-series anomaly detection with applications in smart meter systems and power grids.

Neighborhood Feature Aggregation for Remote Sensing 2025

Proposed neighborhood-based feature aggregation (pooling) for remote sensing image classification, targeting agricultural and environmental monitoring.

Activities and Service

Peer Review: LXAI-NeurIPS 2025; IEEE Robotics and Automation Letters (RA-L); ICRA 2026; ACC 2025; CCTA 2025

Student Mentoring: Chloe Tile; Sophia Martinez-Badillo; Michael Morse; Amran Kassaye; Nazar Oladepo

Awards and Scholarships

Top rank in Iranian National Entrance Exam for Master's programs (Awarded)

PlantCV Workshop + NAPPN 2026 Travel Award 2026
